BIOMETRIC BLOOD GROUP IDENTIFICATION SYSTEM

Final Year B.E. Major Project • Deep Learning AI Analysis Report

RESEARCH / PROTOTYPE SYSTEM DISCLAIMER: This system is a research/prototype fingerprint-based blood-group classification system. It is not clinically validated and must not be used as a substitute for standard serological blood-group testing or for medical/transfusion decisions. All predictions and advisory indicators are provided solely for evaluation purposes.

1. EXAMINATION & SAMPLE AUDIT DETAILS

Report ID:	BG-000579	**Timestamp:**	2026-09-08 19:59:14
Input Source:	MFS100 (Mock Mode)	**Evaluation Status:**	**LOW_CONFIDENCE**
Model Name:	Champion Ensemble V2 (ResNet50 58% + VGG16 38% + EffNet 4%)	**Confidence:**	**53.60%**
Advisory Status:	**NEEDS CONFIRMATION** (τ=70%)	**Advisory Reason:**	**LOW_CONFIDENCE_AND_RH_UNCERTAINTY**

2. BIOMETRIC IMPRESSION & PREDICTED BLOOD GROUP



Predicted Blood Group:

O-

Confidence Score: **53.60%**

Advisory Threshold: **70%**

Advisory Screening: **NEEDS CONFIRMATION**

Derived ABO Group: **O** (89.8% prob) | Rh Factor: **-**

RESEARCH ADVISORY: Advisory threshold passed (≥ 70.0%). Note: Confidence OK does not constitute medical verification. Standard serological testing required for medical/transfusion use.

Rh FACTOR UNCERTAINTY ADVISORY: Rh factor (±) determination is uncertain (narrow Rh probability margin (17.4%); moderate 8-class confidence (53.6%); predicted Rh-negative requires serological validation due to weak ridge-Rh correlation). ABO group appears to be 'O' (89.8% confidence). Standard serological laboratory blood typing is required to confirm Rh factor.

3. 8-CLASS BLOOD GROUP PROBABILITY DISTRIBUTION

A+	A-	AB+	AB-	B+	B-	O+	O-
1.7%	3.9%	0.8%	3.8%	0.1%	0.0%	36.2%	53.6%